The Open Source Firefighter

Public Roadmap

v1.0 · August 2026 · Living document — revised at each release *Document ID:* ***OSF-3.2*** *· one document of the Open Source Firefighter specification family.*

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This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

This document describes where the specification family goes next: what is settled, what is being built, and what the community is expected to carry. It is a direction of travel rather than a schedule — the field and the fire season set the pace.

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1. Specification Core

The coordination system's defining documents.

- **Settled.** Main specification (architecture, coordination protocol, incident command interface, cryptographic event log)
- **Settled.** Layered architecture, event interface and message enumeration, command broadcast, single command surface, backhaul and gateway nodes, expeditionary deployment, governance charter
- **Settled.** Specification assembled and complete
- **Ahead.** Variation taxonomy (systematic capability-axis enumeration; publication-day companion)
- **Ahead.** Front-matter hardening pass (deterrents and category boundaries stated summarization-proof, per red-team findings)

2. Hardware Annex A

The buildable system, three enclosure classes, one architecture.

- **Settled.** Three-class family (Residential / Government-Commercial / Tactical), common architecture, sequencing behavior, self-calibration
- **Settled.** Power, GNSS, radio/network, gateway roles, spectrum posture, export-control boundary
- **Settled.** Coverage engineering: wetting classes (NFDRS-anchored), cycle equation, weather source hierarchy, application rates
- **Settled.** Fasteners, mounting, grounding, wiring palettes, enclosure water management
- **Settled.** Tactical doctrine: weight/carry, joint inventory, self-test, prepositioning, satellite emergency module
- **In progress.** Continuous revision from review findings (continuous)
- **Ahead.** Reference layout figures (see Workstream 5)
- **Ahead.** Freeze/climate appendix expansion; marine environment notes consolidation

3. Protocol & Firmware

The wire format and the code that speaks it.

- **Settled.** Wire format and vocabulary: envelope, byte budget, closed two-dozen-verb vocabulary, quantization rules, capability bits
- **Ahead.** Wire format: field layouts per verb, worked byte-level examples
- **Ahead.** Reference firmware: sequencer + self-calibration (controller)
- **Ahead.** Reference firmware: router/publication policy + mesh coordination
- **Ahead.** Voice annunciation manifest (spec artifact; translations as community contributions) and guided-setup state machine

- **Ahead.** Ledger/snapshot implementation and witness tooling
- **Ahead.** Incident command / TAK (Cursor-on-Target) publication capability

4. Documentation & Guides

Everything a builder, installer, inspector, or council needs.

- **Settled.** Public overview (plain-language, full program)
- **Settled.** Community adoption briefing (councils, districts, departments)
- **Settled.** Supply & Coverage Planning Table (OSF-2 Appendix A) (one-page conservative hydraulics)
- **Settled.** Inspector's Review Draft (code-format shall-paragraphs;)
- **Settled.** Design philosophy (the fifteen rules)
- **Ahead.** Workmanship & Best Practices guide (ferrules, dressing, labeling, wire palettes — "not required; expected in quality work")
- **Ahead.** Guided-assembly sequence (numbered build order per class)
- **Ahead.** Code-Format Edition (full NFPA-patterned rendering; post-publication)
- **Ahead.** Commissioning & annual test checklist (printable, lives in the door pocket)

5. Reference Build Program

Patterns, not products. Process-agnostic, multi-factory by construction.

- **In progress.** First bench infrastructure exists (electronics review/education bench)
- **Ahead.** Four-vendor differential layout pass → convergence table
- **Ahead.** Reference layout drawings per class (manufacturer CAD, dimensioned; double as spec figures)
- **Ahead.** Pattern library: valve brackets, tactical insert set, panel plates, compartment divider — geometry first, printable derivatives first-class
- **Ahead.** Open panel-planning utility (browser-based fit/BOM calculator, no account, no server dependency)
- **Ahead.** Bench prototype, Residential class (first full build against the documents)
- **Ahead.** Review station protocol (community builds graded against checklist; findings to change log)

6. Governance & Program

How the standard stays open, forever.

- **Settled.** Governance charter (six clauses + operators-prevail; Clauses 2 and 7 unweakenable)
- **Settled.** Message-category enumeration as the mission boundary (structural, not policy)
- **Settled.** Equipment category and certification pathway declared (industrial control panel class; UL 508A route; not a life-safety listing regime)
- **In progress.** Intellectual property protection and permanent-open-release instruments (in preparation; the standard publishes free and stays free)
- **Ahead.** Public repository and project site (git-backed; commit history as the change log)
- **Ahead.** Proposal/disposition process stood up in public
- **Ahead.** Steward succession and continuity documentation

7. Validation & Testing

Prove it, in public, with the instrument built into every box.

- **Settled.** Adversarial review round 1 (hostile-reader analysis; findings dispositioned into the documents)
- **In progress.** Inspector review (installation requirements under review by a career building inspector)
- **Ahead.** Adversarial review at document freeze (real documents, multiple models, hostile personas)
- **Ahead.** Bench hydraulic validation (self-calibration against real supplies; planning-table verification)
- **Ahead.** Fold test (tactical case abuse trial by the intended population)
- **Ahead.** Pilot street (first coordinated multi-home deployment; ledger data published)

- **Ahead.** Empirical re-wet tables (wetted-surface persistence by material, from fleet ledger data)

8. Community & Adoption

The standard belongs to the people it protects.

- **Settled.** Activation instruments written (overview + adoption briefing)
- **Settled.** Dedication (Rifle Helitack crew, Knowles Fire, June 27, 2026)
- **Ahead.** Publication day release (M1)
- **Ahead.** Activation outreach: Firewise communities, extension services, VFD networks, maker/trade channels
- **Ahead.** Contribution intake (war stories welcome; the record learns to hear practitioners)
- **Ahead.** Configuration library seeded (first two or three known-good configurations — see §9)
- **Ahead.** Build registry open (what people built, what they substituted, what broke)
- **Ahead.** Safety-defect reporting channel published
- **Ahead.** Model council resolution / permitting recognition template
- **Ahead.** First adopting jurisdiction; first insurance-recognition conversation
- **Ahead.** Translations of voice manifest and public documents

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9. The Development Commons — what has to exist for the community to work

The specification can be finished by one person. The thing that makes it *usable* cannot be, and it is worth naming precisely what the community is for, because “build a community” is not a plan.

The goal: known-good configurations

The target state is that installing a node on a common structure is **trivial** — not because the system got simpler, but because someone already worked out that configuration, published it, and a hundred people refined it.

The precedent worth studying is the open engine-management community. An open ECU is a general-purpose device and, on its own, an expert-only proposition: you would need to derive fueling and timing from theory for your specific engine. What made it ordinary was not the hardware. It was that the community produced and maintained **per-platform base configurations** — a tested starting map for a Miata, for a Civic, for whatever you had. A newcomer stopped starting from theory and started from something that already ran, then tuned from there. Strapping one to a common car became a weekend job that thousands of people had already documented.

That is exactly what this project needs, and the axis is structure and site rather than engine. The configurations the community should converge on look like:

- Single-wide manufactured home, municipal supply, hose-bib class
- Ranch house on well and cistern, with pump control
- Hillside home, gravity-fed tank, no pump
- Manufactured-home park — multiple nodes on a shared supply, coordinated
- Outbuilding, barn, or detached shop
- Expeditionary case, tactical class
- Utility right-of-way corridor segment

with regional variants layered on top, because radio allocation, freeze exposure, and water-supply practice all differ by jurisdiction.

What a known-good configuration contains

A configuration is a citable artifact, not folklore. Each one carries:

- **The archetype and its assumptions** — roof geometry, supply class, climate, jurisdiction
- **Coverage layout** — head placement and count, zone division
- **Supply math** — worked against the Planning Table (OSF-2, Appendix A)
- **Parts list at a price tier**, with the substitutions someone has actually verified
- **Panel layout and wiring variant**

- **Firmware configuration** — zone count, timings, regional radio parameters
- **Its evidence** — who built it, what was tested, what failed, and any thermal-verification pass
- **Known gotchas** — the things that bit the people who built it

Two properties are non-negotiable. **Configurations are versioned and attributed**, so a builder knows what they are trusting and can go ask the person who wrote it. And **configurations carry their test status honestly**, including "untested" — because the real hazard here is not a missing configuration, it is a plausible-looking one that under-waters and gives a homeowner false confidence. A configuration that has never been run must say so in the same breath as its parts list. Thermal-signature verification (Annex A) is the field check that closes this loop: a configuration is proven when the roof reads wet.

The first two or three configurations are seeded by the author. **Every one after that is the community's**, which is the point — the same substitution-permitted doctrine the parts catalog already runs on, applied to whole installations.

The infrastructure that produces them

Functions, not products — platforms change, these do not:

- **A canonical record that no single company owns.** The documents are plain-text and version-controlled already, which is most of the work. Mirror it: Charter Clause 2 says the *system* carries no mandatory external dependency, and the same discipline applies to the project. If a forge disappears, the standard must not.
- **A change-proposal process.** The most important item on this list, because interoperability is the product and the charter routes vocabulary additions through governance. It needs a template, a discussion window, and a decisions-of-record log. Without one, "governance" is one person answering threads.
- **An archived asynchronous forum.** The record. Real-time chat is for velocity and is ephemeral by design; a decision that lives only in chat did not happen.
- **A build registry.** What people built, what they substituted, what broke. This is the feedback loop that fills in both the parts catalog and the configuration library — and it is the item most easily forgotten, because it has no software analogue.
- **A conformance suite and a way to claim conformance.** This is the anti-fork instrument: it defines what *is* an OSF node, and it makes contributing easier than forking.
- **A private safety-defect channel** with coordinated disclosure. Most projects add one after an incident. Safety-adjacent hardware does not get that grace.
- **Regional working groups**, because compliance, climate, and supply practice are local and no central maintainer can hold them all.

The name as the conformance instrument

The design is free. The **name** is the thing that carries a promise, and it is the only practical lever for keeping that promise honest.

This is the structure mature open-hardware projects converge on, and it separates two things that are usually confused. The specification, the drawings, and the reference implementation are open under their licences: anyone may build, modify, sell, and fork without asking. The **mark** — the project name and its identifying logo — is held, and its use is granted on stated terms. A builder who meets the conformance suite may say the thing is an OSF node. A builder who does not may still build, sell, and ship whatever they like; they simply may not call it that.

Why this matters more here than in most projects: a non-conforming node in a coordination domain does not merely underperform, it **draws water out of turn**. A device that ignores the schedule and calls itself compliant is a hazard to every neighbour on the supply. Conformance is a safety property, and a conformance claim needs something behind it.

Two provisions distinguish a workable mark from a hollow one:

- **The grant is objective, not discretionary.** Meet the published conformance suite and the grant follows; fail it and the grant lapses. It is not a relationship to be managed, a fee to be paid, or a favour to be granted. Discretionary marks become gatekeeping, and gatekeeping is the failure this project is built to avoid.
- **The grant is not revocable at will.** Revocable-at-any-time marks are the same failure as revocable patent pledges: they look like a commitment and function as leverage. A conformance grant that can be withdrawn without cause is worth nothing to the builder relying on it.

A secondary mark for compatibility — *works with*, *interoperates with* — costs nothing to offer and lets adjacent

projects state a relationship without claiming conformance they have not tested for. Established projects in this space do exactly that, and it defuses most of the friction a mark would otherwise create.

What this is not: a licence fee, an approval queue, an endorsement, or a warranty. It is a claim about testable behaviour, and nothing more.

Roles, decision rights, and succession

Written down, publicly, in ordinary language: who decides, how a contributor becomes a maintainer, how disputes resolve, and what happens when a maintainer — including the author — is unavailable. Founder-led projects most reliably fail at the transition from founder to community, and the failure is social rather than technical.

The succession question and the estate instrument are the same problem, and belong in the same conversation with counsel: the instrument that guarantees the work stays open should also say who carries it.

A code of conduct belongs here too, and is best understood as a safety brief rather than an administrative formality — the same document type as the briefing given before anyone steps onto a live range.

Licensing — three decisions of record

Hardware, documentation, and software are different license families and need separate, deliberate choices, recorded with their reasoning. Ambiguity here is the most common reason a willing contributor walks away.

What this is not

None of this gets pre-built. An empty forum is worse than no forum, and infrastructure without people is a maintenance burden that signals abandonment. **This section is a specification for what to stand up when there is traffic to justify it** — in roughly the order listed, since the record and the change process must exist before the first outside contribution arrives, and everything else can wait for demand.

Where this goes

- The twelve-month development window (M2) exists precisely because the remaining work — firmware, builds, testing, adoption — is designed to be done in the open, by many hands, against a published standard. Everything on this list is buildable by strangers from the documents; that is the point of the documents.*